



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 20: Monopoly

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 20

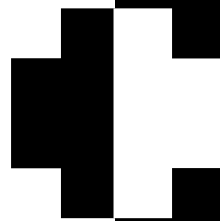
Last lecture, we studied **Welfare Analysis**

- Revised an example of PPF
- Started Welfare Economics and Economic Efficiency

In today's lecture, we will start a new chapter **Market Failures:**

- We will start with **Monopoly**
 - Monopoly definition
 - Barriers to entry
 - Monopoly's profits

**Consumer
Theory**



**Producer
Theory**

Welfare Economics and Competitive Markets

Market Failures

Game Theory

Market Failures

Microeconomics

Departing from Competitive Assumptions

Factors that may distort the ability of competitive markets to achieve efficiency:

1. **Imperfect competition:** All situations in which economic agents exert some market power in determining price.
2. **Externalities:** Competitive price system also fails to allocate resources efficiently when there are interactions among firms and individuals that are not adequately reflected in the market prices → difference between private and social marginal costs!
3. **Public Goods:** goods that are nonrival and nonexclusive → individuals adopt a free rider stance
4. **Imperfect information:** If economic actors are uncertain about prices, there is no reason to expect that the efficiency property of competitive pricing will be retained

Departing from Competitive Assumptions

Factors that may distort the ability of competitive markets to achieve efficiency:

1. **Imperfect competition:** All situations in which economic agents exert some market power in determining price
2. **Externalities:** Competitive price system also fails to allocate resources efficiently when there are interactions among firms and individuals that are not adequately reflected in the market prices → difference between private and social marginal costs!
3. **Public Goods:** goods that are nonrival and nonexclusive → individuals adopt a free rider stance
4. **Imperfect information:** If economic actors are uncertain about prices, there is no reason to expect that the efficiency property of competitive pricing will be retained

Monopoly definition

Market Failures

Homework Lecture 20

Watch the video and answer the following questions about it:

<http://www.youtube.com/watch?v=7UWgKZsKZOc>

1. Which examples of barriers to entry are mentioned in the video?
2. Why monopolies are seen as the “bad guys”?

1. Market Failures: Monopoly

What is a Monopoly?

- A **single firm** that serves the market
- This firm **faces the entire demand curve** for its output, it has no close substitutes
- The **monopolist knows** this, so he **makes a decision on how much to produce** and will determine the good's price

Why does a monopoly exist?

- Other firms find it unprofitable or impossible to enter the market

→ **Source of monopoly power: Barriers to entry**

Monopoly: Barriers to Entry

Market Failures

2. Monopoly: Barriers to Entry

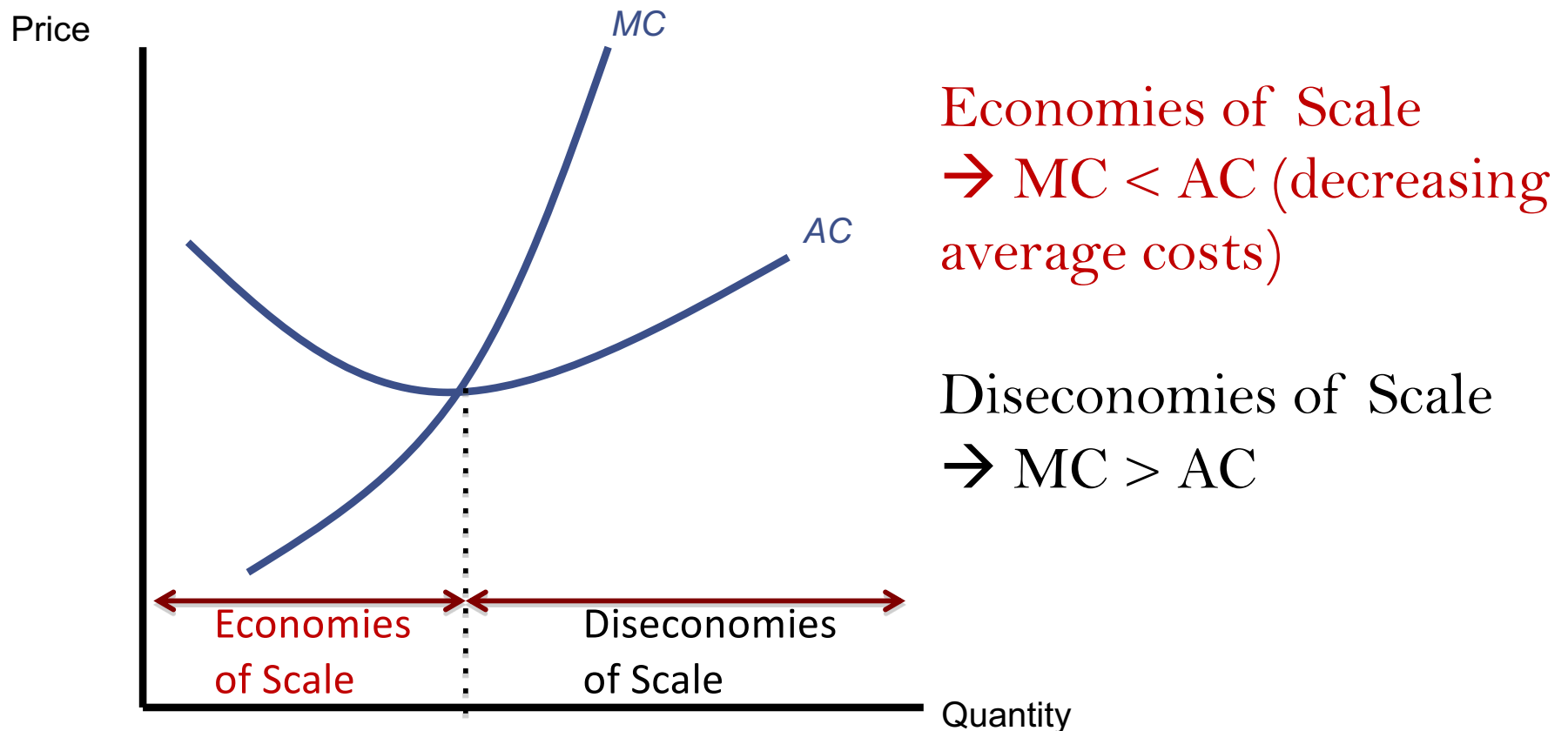
There are two types of barriers to entry:

- Technical Barriers
- Legal Barriers

2. Monopoly: Barriers to Entry

Technical barriers to entry

1. Production of the good exhibit decreasing average costs over a wide range of output levels



2. Monopoly: Barriers to Entry

Technical barriers to entry

1. Production of the good exhibit decreasing average costs over a wide range of output levels
 - relatively large-scale firms are low-cost producers
 - Commonly referred to as a **natural monopoly**

Once a monopoly is established, entry will be difficult because any new firm must produce relatively low output levels → with relatively high average costs.

2. Special knowledge of a low-cost productive technique
3. Ownership of unique resources (mineral deposits, land locations, unique managerial talents)



Author: Vicente Izurieta
(I-2020)

2. Monopoly: Barriers to Entry

Legal barriers to entry

1. Legal protection of a product by a patent or copyright

Profitable products are shielded (for a time) from the direct competition by potential imitators, where the basic technology for these products is assigned to one firm

Why do governments do this?

Makes innovation more profitable → incentive to promote innovation

2. Awarding of an exclusive franchise to serve a market

Public utility services (gas, electric), communication services, post office, etc

Why do governments do this?

The industry in question is a natural monopoly (although sometimes franchises are based on a political rationale)



Created in imageflip.com and recovered from: <https://imgflip.com/i/45pg70?herp=1592596122177>

Author: Ignacio Muñoz
(I-2020)

Monopoly: Profit Maximization and Output Choice

Market Failures

3. Monopoly: Profit Maximization and Output Choice

Monopolist will maximize profits:

$$\text{Max}_{\{Q\}} \pi = PQ - C(Q)$$

First order condition:

$$\frac{\partial \pi}{\partial Q} = P + Q \left(\frac{\partial P}{\partial Q} \right) - MC = 0$$

$$\boxed{MR = P + Q \left(\frac{\partial P}{\partial Q} \right) = MC}$$



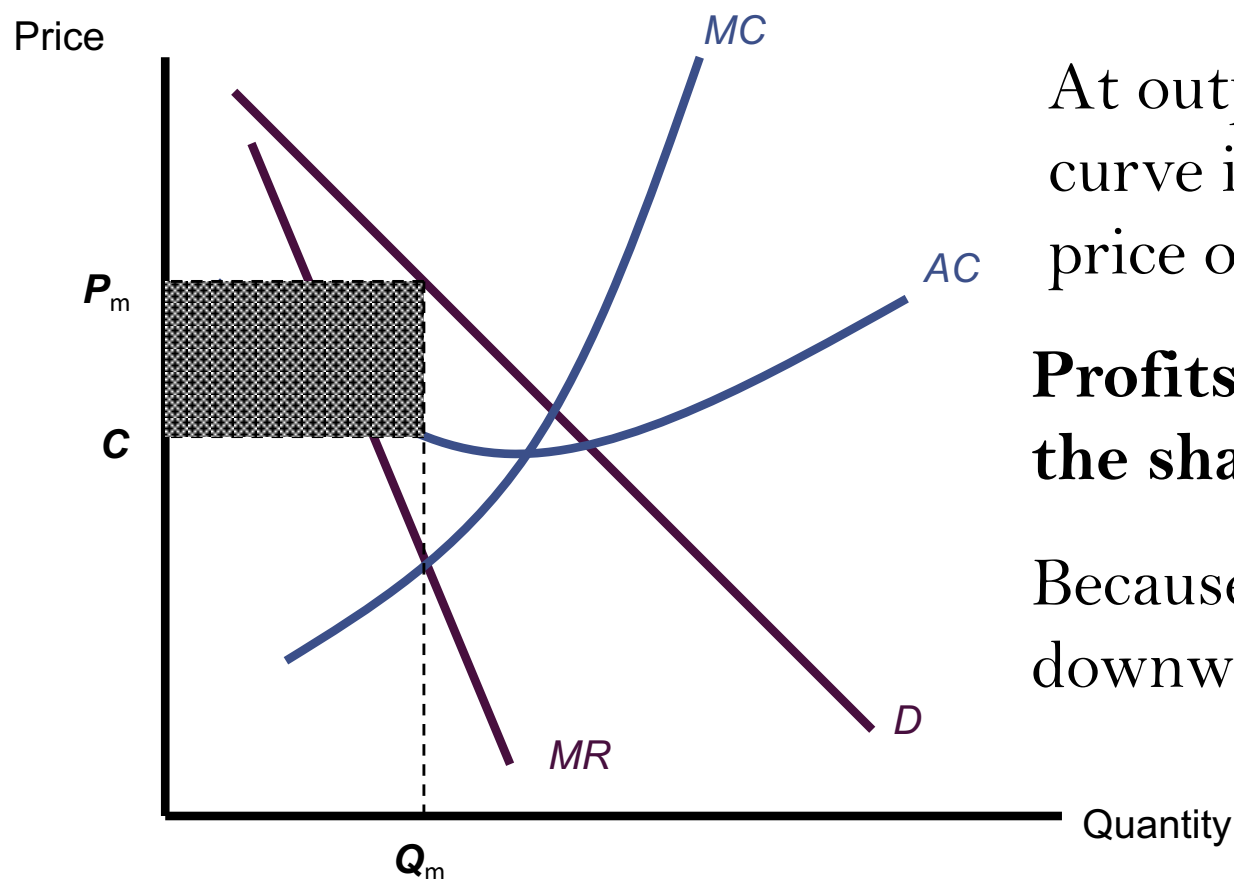
Monopolist knows that he faces the market demand curve, **price is NOT fixed!**

Second order condition:

$$\frac{\partial^2 \pi}{\partial Q^2} < 0 \rightarrow \frac{\partial MR}{\partial Q} < \frac{\partial CM}{\partial Q}$$

3. Monopoly: Profit Maximization and Output Choice

Monopolist maximizes profits $\rightarrow MR = MC \rightarrow$ finds Q_m



At output level Q_m , demand curve indicates that a market price of P_m will prevail

Profits can be found in the shaded rectangle

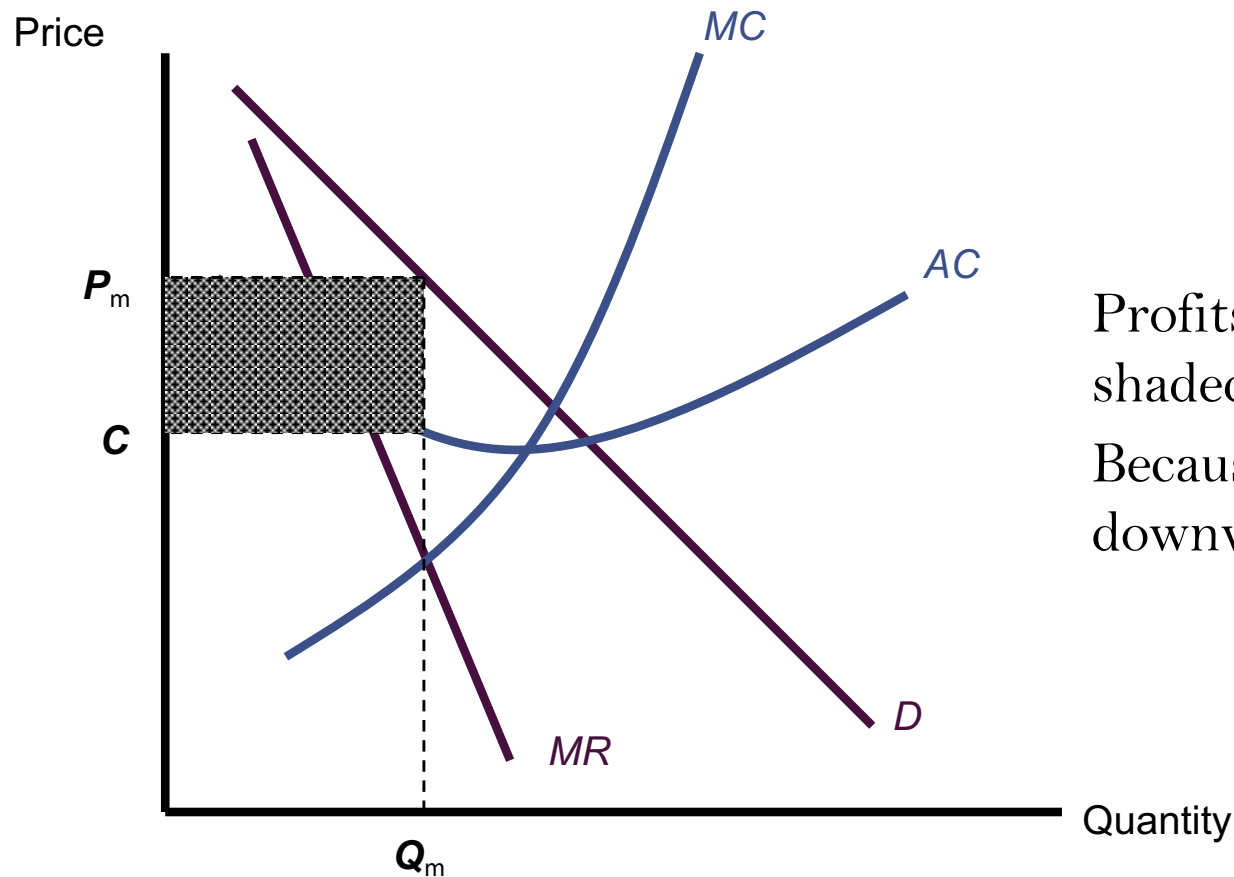
Because demand curve is downward-sloping, $P_m > MC$

3. Monopoly: Profit Maximization and Output Choice

Monopolist's profits will be positive as long as $P > AC$

This profits can continue in the Long-run because entry is not possible

Some economists refer to this profits in the LR as “*monopoly rents*”



Profits can be found in the shaded rectangle

Because demand curve is downward-sloping, $P_m > MC$

WHEN YOU REALIZE THAT IF YOU CREATE A
MONOPOLY YOU CAN SET THE PRICE YOU WANT
AND PEOPLE HAVE TO PAY IT



Example

The demand curve is:

$$Q = 24 - P$$

And the costs of a monopoly producer are given by:

$$C(Q) = Q^2 + 12$$

- a) Find the marginal revenue curve
- b) Find the level of output that maximises the monopolist's profits.
- c) Find the profits and graph the profits and revenues in terms of the quantity produced.

Example: Solution

The demand curve is:

$$Q = 24 - P$$

And the costs of a monopoly producer are given by:

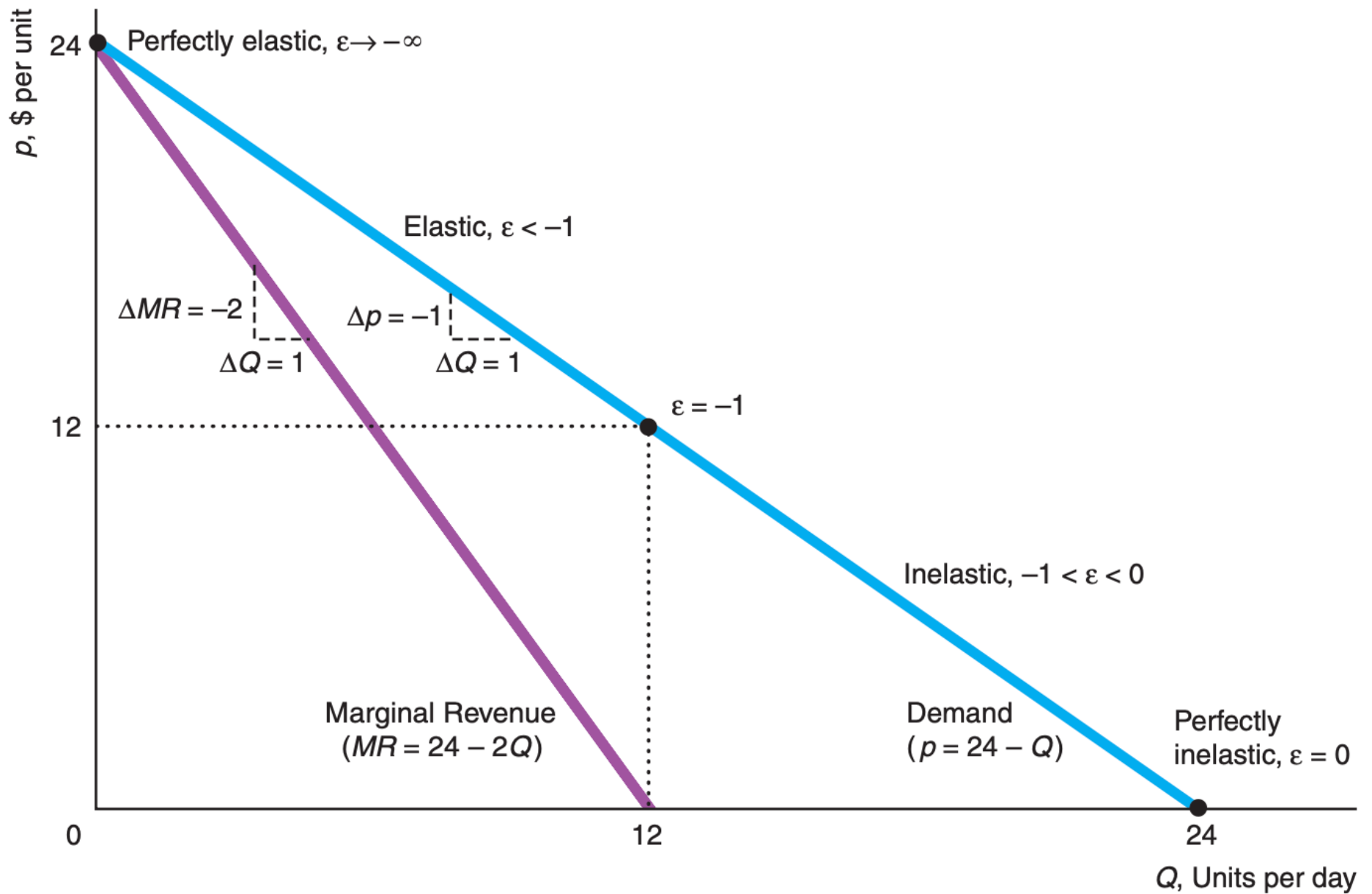
$$C(Q) = Q^2 + 12$$

a) Find the marginal revenue curve

The revenue is:

$$\text{Revenue} = P \cdot Q = (24 - Q) \cdot Q = 24Q - Q^2$$

$$\text{Marginal Revenue} = \frac{\partial(24Q - Q^2)}{\partial Q} = 24 - 2Q$$



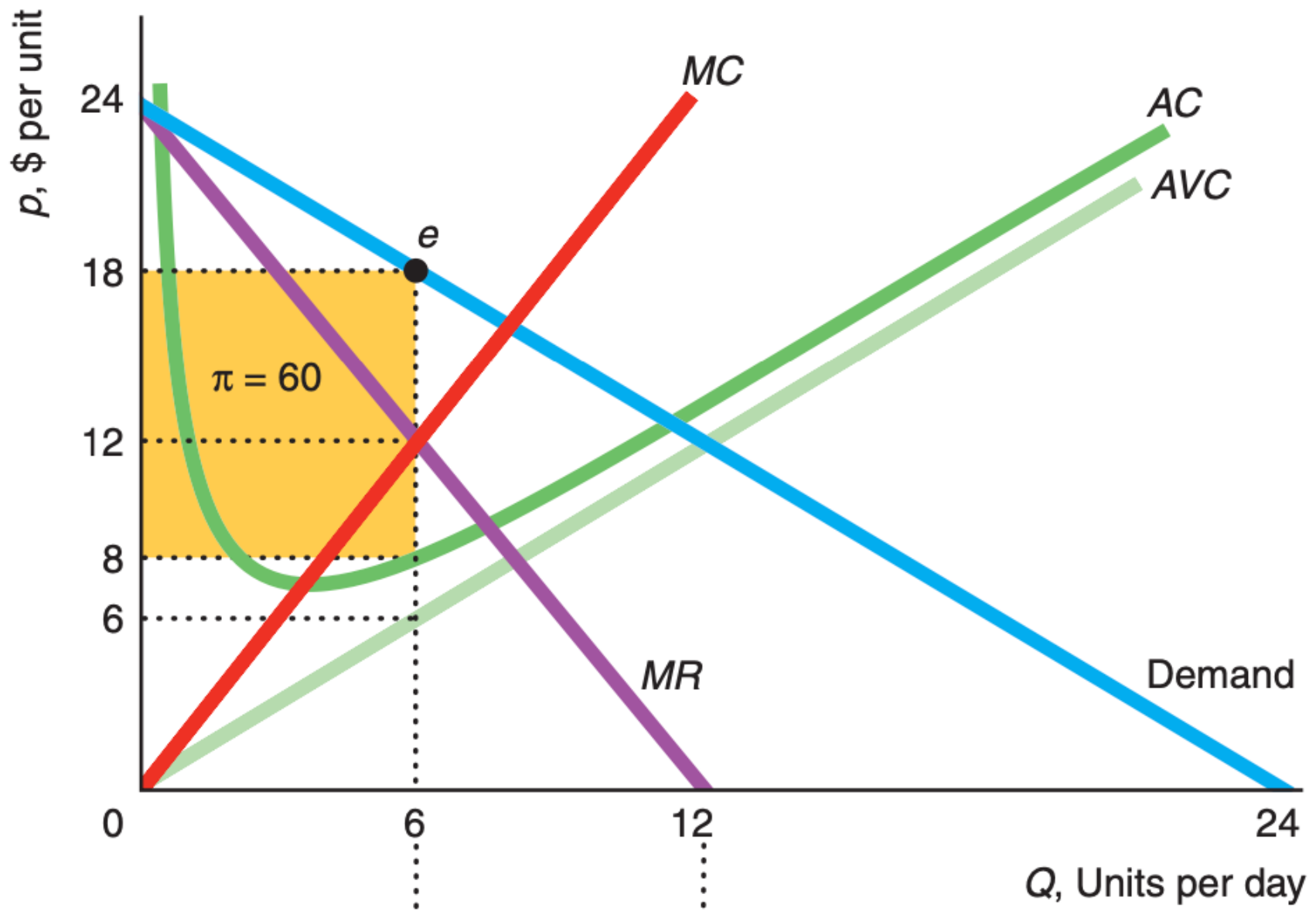
Example: Solution

- b) Find the level of output that maximises the monopolist's profits.

The monopolist maximises profits when the marginal cost equals the marginal revenue:

$$\begin{aligned} MC &= MR \\ 2Q &= 24 - 2Q \\ Q_m &= \frac{24}{4} = 6 \end{aligned}$$

(a) Monopolized Market



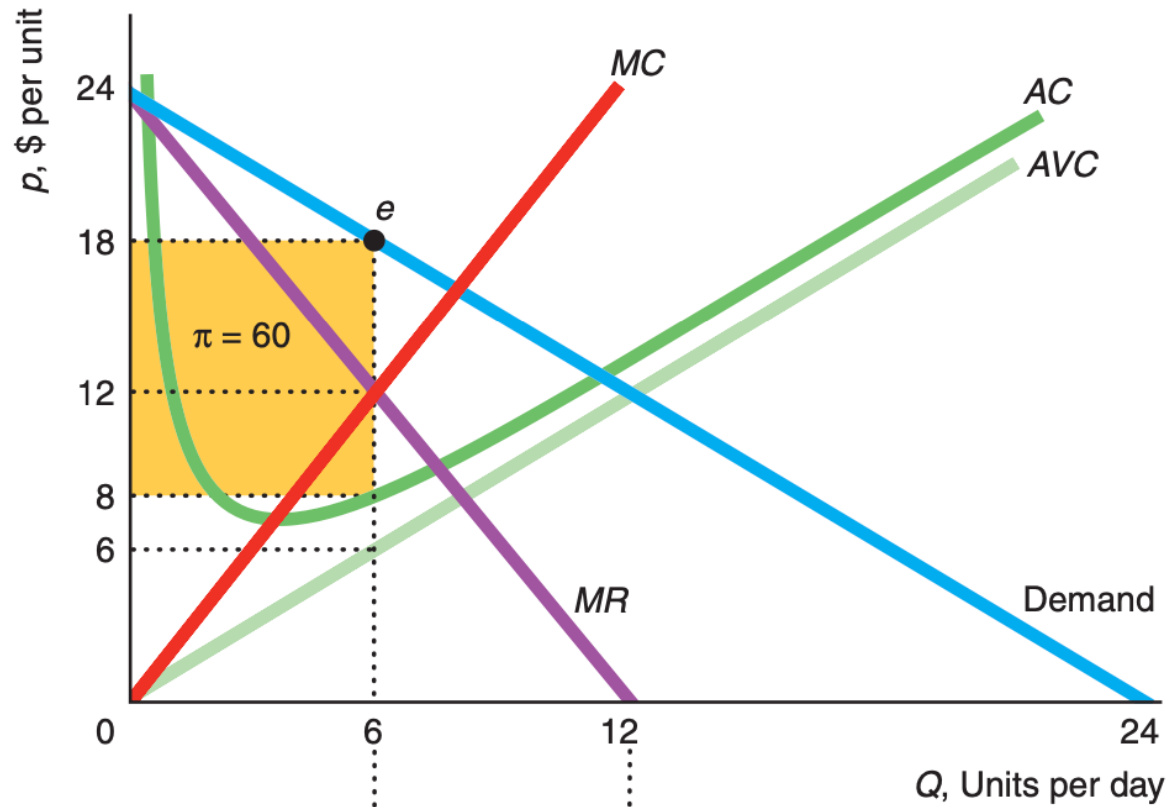
Example: Solution

- c) Find the profits and graph the profits and revenues in terms of the quantity produced.

If $Q_m = 6$, the profits are:

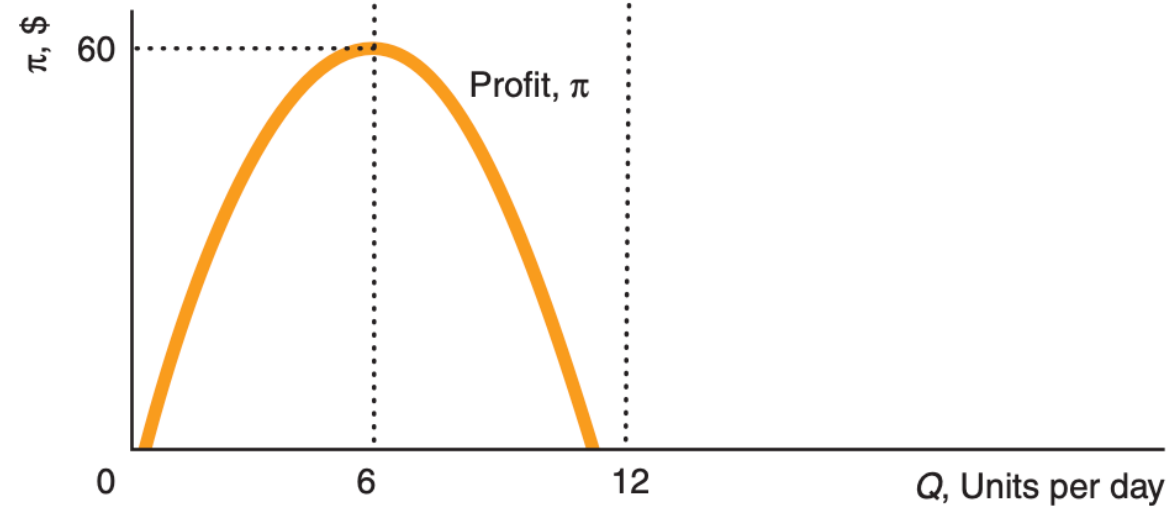
$$\pi = P \cdot Q - C(Q) = 18 \cdot 6 - (6^2 + 12) = 60$$

(a) Monopolized Market



Perloff (2015)

(b) Profit



Activity 1

Suppose the market for Olympic-quality Frisbees (Q , measured in Frisbees bought per year) has a linear demand curve of the form:

$$Q = 2,000 - 20P$$

And the costs of a monopoly Frisbee producer are given by:

$$C(Q) = 0.05Q^2 + 10,000$$

- a) Find the level of output that maximises the monopolist's profits.
- b) Find the maximum level of profits.

3. Monopoly: Profit Maximization and Output Choice

Monopoly: Rule of the thumb for pricing

We know that price and output should be chosen so that $MR = MC$, but how can the manager of a firm find the correct price and output level in practice?

Remember, most managers have only limited knowledge of the average and marginal revenues curves that their firms face

$$MR = \frac{\Delta R}{\Delta Q} = \frac{\Delta(PQ)}{\Delta Q} = P + Q \frac{\Delta P}{\Delta Q}$$

$$MR = P + P \left(\frac{Q}{P} \right) \left(\frac{\Delta P}{\Delta Q} \right)$$

$$MR = P + P \left(\frac{1}{\varepsilon_d} \right) = MC$$

$$\left(\frac{1}{\varepsilon_d} \right) = \frac{MC - P}{P} \rightarrow \frac{P - MC}{P} = - \left(\frac{1}{\varepsilon_d} \right)$$

3. Monopoly: Profit Maximization and Output Choice

Monopoly: Rule of the thumb for pricing

$$\frac{P - MC}{P} = -\left(\frac{1}{\varepsilon_d}\right)$$

This relationship provides a rule of the thumb for pricing
(P-MC)/P is the **markup** over marginal cost as a percentage of price and depends inversely on the elasticity of demand

Pure monopoly is rare, but markets with several firms are more common. This expression also hints the market power (**Lerner Index of Monopoly Power**):

$$L = \frac{P - MC}{P} = -\left(\frac{1}{\varepsilon_d}\right)$$

Activity 2

Find in which regions of the market demand curve (elastic/inelastic) the monopolist will choose to operate:
(Hint: test the sign of the MR first)

- a. Elastic
- b. Inelastic

$$\frac{P - MC}{P} = -\left(\frac{1}{\varepsilon_d}\right)$$

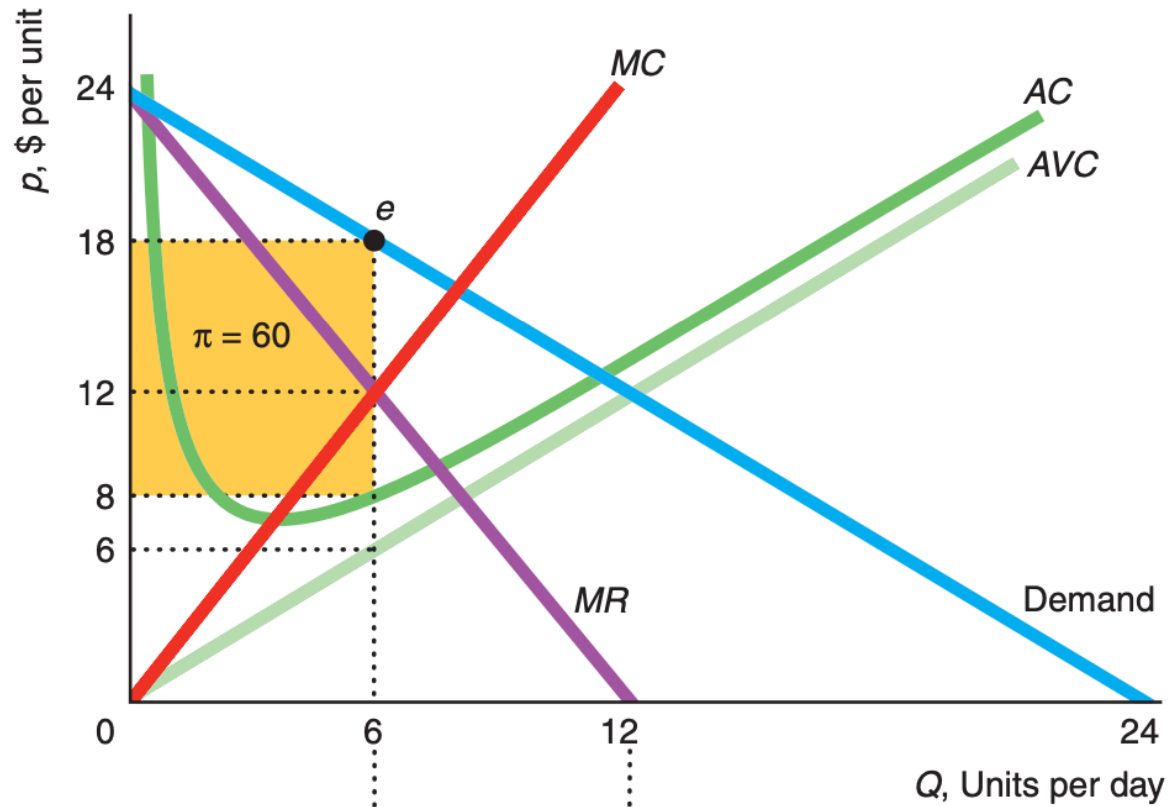
Código de evento Wooclap

RIHFAB

<https://app.wooclap.com/RIHFAB?from=instruction-slide>

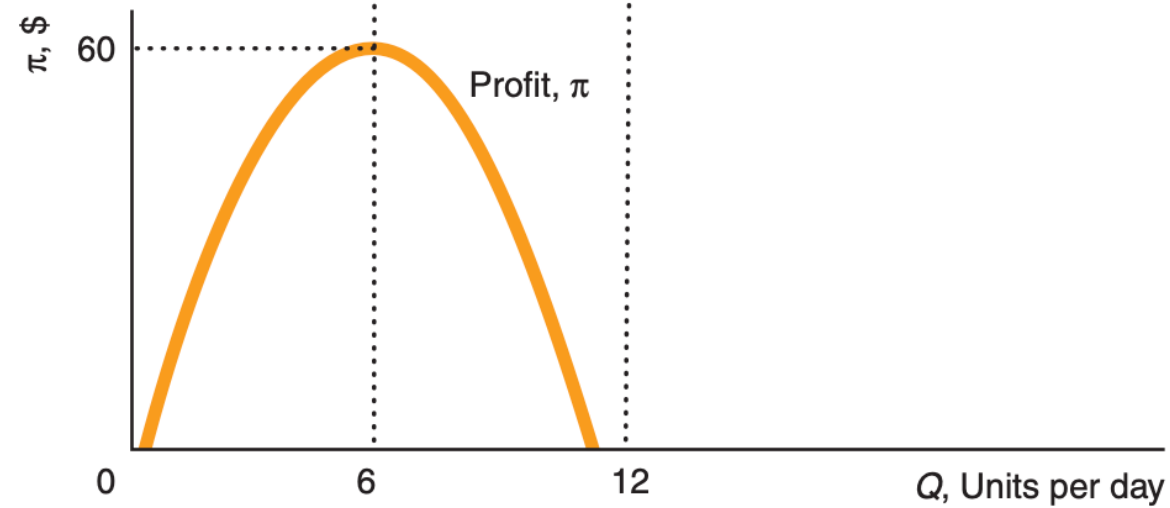


(a) Monopolized Market



Perloff (2015)

(b) Profit



3. Monopoly: Profit Maximization and Output Choice

Is there a monopoly supply curve?

Answer: No

- When we studied perfectly competitive markets, we talked about an industry supply curve
- We constructed the long-run supply curve by allowing the market demand to shift and observing the supply curve that was traced out by the series of equilibrium price-quantity combinations
- With a fixed demand curve $\rightarrow$ supply “curve” will be one point, when $MC=MR$. If demand curve shifts $\rightarrow$ MR shifts $\rightarrow$ new profit-maximizing output will be chosen
- Connecting the resulting series of equilibrium points would have little meaning (and the locus might have a very strange shape)

Monopoly: Resource Allocation

Market Failures

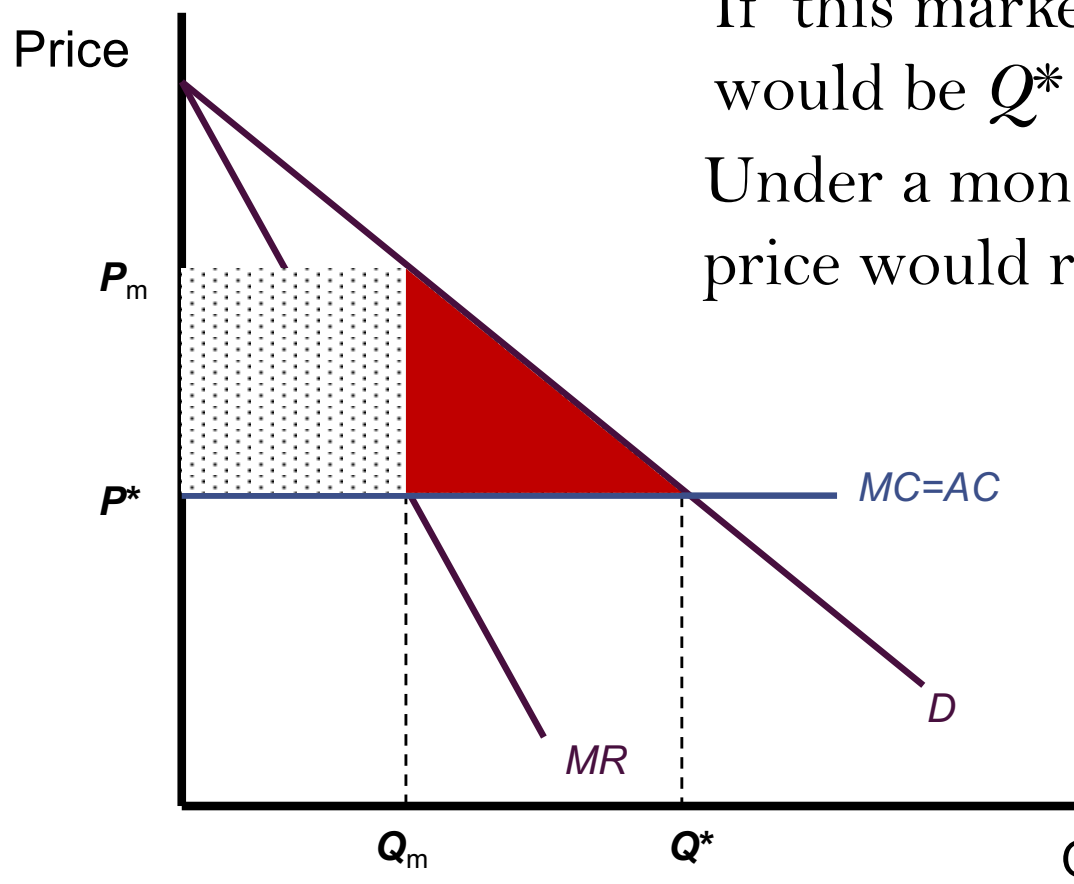
4. Monopoly: Resource Allocation

In a monopoly, market price no longer conveys accurate information about production costs $\rightarrow$ resources will be misallocated

Comparison between monopoly and a competitive market (constant cost industry)

If this market was competitive, output would be Q^* and price would be P^*

Under a monopoly, output would be Q_m and price would rise to $P_m \rightarrow P_m > P^*, Q_m < Q$



Consumer surplus will fall

Producer surplus will rise

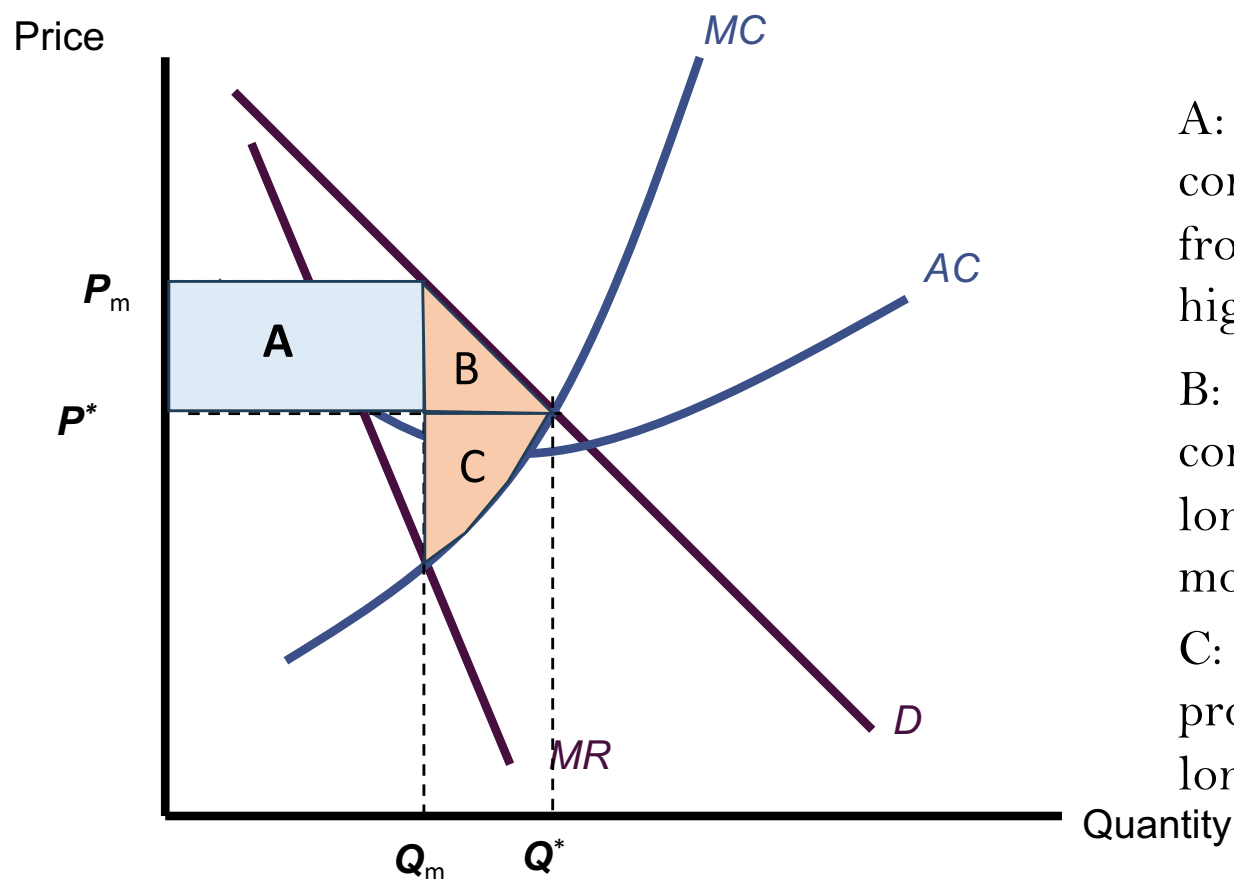
There is a deadweight loss from monopoly

Graphs by South-Western, a division of Thomson Learning. All rights reserved.

4. Monopoly: Resource Allocation

In a monopoly, market price no longer conveys accurate information about production costs $\rightarrow$ resources will be misallocated

Comparison between monopoly and a competitive market



A: is a transference from the consumers to the producers from those units consumed at a higher price

B: is a **deadweight loss** for the consumers, for those units no longer consumed in the monopoly

C: is a **deadweight loss** for the producers, for those units no longer sold in the monopoly

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.